

Solid State Optics

Lukas Hein, Elias Schwarzkopf

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the refractive index, extinction coefficient, absorption coefficient, bandgap, and pulse matrix element for all given semiconductor samples.

1 Introduction

Fourier Transform Infrared (FTIR) spectroscopy is a powerful method for determining the electronic properties of metals and semiconductors. The FTIR technique is based on the interaction of infrared light with the material. Reflection and transmission are used to determine important electronic properties. Especially for semiconductors, the FTIR method is particularly useful because it helps to adjust parameters such as the doping concentration or the bandgap. For building applications like LEDs, transistors, or solar cells, precise knowledge of the bandgap and doping concentrations is required. In contrast to other common methods, such as spectroscopy with a monochromator, the FTIR method is much faster and more precise. Instead of varying the wavelength through multiple measurements, the FTIR method can measure the entire spectrum at once due to the use of a Michelson interferometer, which enables the interference of light. After Fourier transformation, the complete spectrum is obtained in a single step.

Firstly, the theoretical basics of light-matter interaction and some important formulas regarding reflection and transmission will be discussed. Following this theoretical introduction, the experimental setup will be described. The final part of this paper will focus on data evaluation, starting with the gas absorption of air. The second part of the data evaluation will involve determining the signal-to-noise ratio. The last part will address the determination of several previously

2 Theory

2.1 Wave equations

The propagation of electromagnetic waves in a medium are described with the wave equations:

$$\nabla^2 \vec{E} = \frac{\mu\epsilon}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} + \sigma\mu_0\mu \frac{\partial \vec{E}}{\partial t} \quad (1)$$

$$\nabla^2 \vec{H} = \frac{\mu\epsilon}{c^2} \frac{\partial^2 \vec{H}}{\partial t^2} + \sigma\mu_0\mu \frac{\partial \vec{H}}{\partial t} \quad (2)$$

with $\vec{E}$ as the electric field vector, $\vec{H}$ as the magnetic field vector, μ as the permeability, ϵ as the permittivity, σ as the conductivity, and c as the speed of light in vacuum.

Under the assumption of plane waves with the components $\vec{E} = E_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$ and $\vec{H} = H_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$ can be used as an ansatz to solve the wave equations ?? and ??. The dispersion relation of the plane waves is then described with:

$$k^2 = \omega^2 \cdot \left(\epsilon_0 + i \frac{\sigma}{\omega \epsilon_0} \right) \quad (3)$$

with k as the wave vector, ω as the angular frequency, and ϵ_0 as the permittivity of vacuum.

2.2 Plane parallel layer

The investigated semiconductor samples can in first approximation be described as a plane parallel layer ?? with a thickness d and refractive index N_2 which acts like a fabry-perot interferometer. Light is sim-

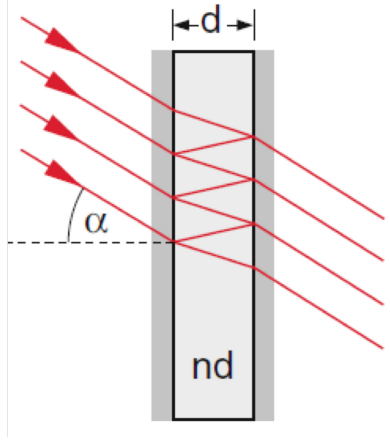


Abbildung 1: Simplified illustration of the measured samples as a plane parallel plate with thickness d and refractive index N_2 . The surrounding air is described with a refractive index $N_1 \approx 1$ [?]

plified as beams. Every incoming beam is partially reflected and partially transmitted at the first interface. The total power reflectivity R_1 is described with:

$$R_1 = \left| \frac{1 - N_2}{1 + N_2} \right|^2 \quad (4)$$

without considering the phase shift of the reflected beam. Like shown in fig. ??, the total power reflectivity of R then consists of a reflection of the incoming beam at the interfaces of the material. Additionally, the absorption of the material is considered. Therefore, R is described with:

$$R = R_{11} + \frac{(1 - R_{11})^2 R_{12} e^{-2d\beta}}{1 - R_{11} R_{12} e^{-2d\beta}} \quad (5)$$

with the absorption coefficient:

$$\beta = \frac{2\kappa\omega}{c}. \quad (6)$$

The total transmission T can be determined in a similar fashion:

$$T = \frac{(1 - R_{11})(1 - R_{12})e^{-d\beta}}{1 - R_{11}R_{12}e^{-2d\beta}}. \quad (7)$$

The assumption that the reflection of the first interface is the same as the reflection of the second interface is in reality not correct. Disturbances like roughness of the interfaces or the absorption of the material lead to a different reflection coefficients of the two interfaces.

Considering this and the phase of the light, the total power reflectivity R is described with:

$$R = \left| \frac{(e^{i\frac{2\omega}{c}N_2d} - 1)(1 - N_2)}{e^{i\frac{2\omega}{c}N_2d}(1 - N_2) - (1 + N_2)} \right|^2. \quad (8)$$

ω is the angular frequency of the light and c the speed of light. The phase shift of the light is described with $e^{i\frac{2\omega}{c}N_2d}$. In contrast to ??, equation ?? shows a periodic modulation of the reflectivity because of interference. The difference between the maxima of the modulation can be calculated with:

$$\Delta\nu = \nu_{m+1} - \nu_m = \frac{1}{2dN_2} \quad (9)$$

2.3 Optical properties

The refractive index N :

$$N(\omega) = n(\omega) + i\kappa(\omega) \quad (10)$$

consists of a real part n and an imaginary part κ and is related to the dielectric function ϵ :

$$\epsilon = N^2 = n^2 - \kappa^2 + 2in\kappa \quad (11)$$

2.4 Drude model

The drude model is a classical model in which the electrons in a solid are treated as a gas of free particles. In this model the dielectric function ϵ can be described as:

$$\epsilon = 1 - \frac{\omega_p^2 \tau^2}{1 + \omega^2 \tau^2} + i \frac{\omega_p^2 \tau}{\omega(1 + \omega^2 \tau^2)} \quad (12)$$

with ω_p as the plasma frequency:

$$\omega_p = \sqrt{\frac{N_e e^2}{\epsilon_0 m^*}} \quad (13)$$

and τ as the relaxation time of the electrons. m^* is the effective mass of the electrons and N_e the charge carrier density. At the plasma frequency ω_p the real part of the refractive index becomes nearly zero $n(\omega_p) \approx 0$ and therefore the phase velocity and wavelength diverge. This means that all electrons are oscillating in phase. Absorption of light is not possible for frequencies below the plasma frequency.

Eq. ?? can be simplified through the introduction of the high frequency conductivity σ

$$\sigma = \frac{N_e e^2 \tau}{m^*} \frac{1}{1 - i\omega\tau} \quad (14)$$

to a expression including the lattice contribution of the material ϵ_{LC} :

$$\epsilon = \epsilon_{LC} + i \frac{\sigma}{\omega \epsilon_0}. \quad (15)$$

For metals the lattice contribution is set to one $\epsilon_{LC} = 1$ as the electrons are considered to be a free electron gas. Using equations ??,?? and ?? with the effective mass $m^* = m_e$ equal to the electron mass, the theoretical reflectivity of a metal can be calculated. The result is shown in fig??. The used constants for the

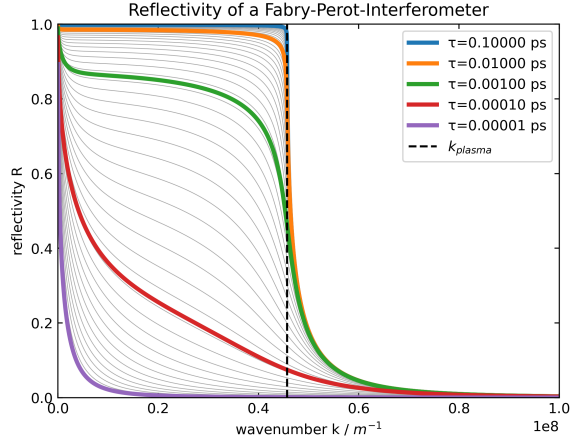


Abbildung 2: Theoretical reflectivity of a metal according to the drude model. The reflectivity is reduced significantly for wavenumbers above k_{plasma}

theoretical reflection of a metal are listed in table ??

charge carrier density N_e	$5.9 \cdot 10^{28} \text{ m}^{-3}$
electron mass m_e	$9.11 \cdot 10^{-31} \text{ kg}$
layer thickness d	$500 \mu\text{m}$

Tabelle 1: Constants used for the calculation of the theoretical reflectivity of a metal, shown in fig??. The charge carrier Density is taken from [?]

For frequencies below the plasma frequency the reflectivity is nearly one. However, if the frequency of the light is larger than the plasma frequency, the reflectivity is significantly reduced.

2.5 Semiconductors

While metals are described properly with the drude model, for semiconductors the bandgap has to be considered additionally. Like shown in ?? the band-

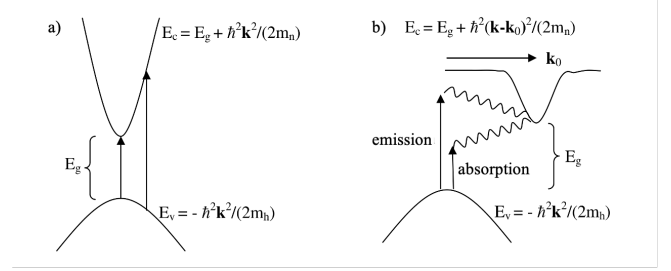


Abbildung 3: a) direct and b) indirect semiconductor bandgap. If a photon is absorbed by a valence electron the excitation to the conduction band is for the indirect bandgap only possible with an additional phonon absorption or emission process.[?]

gap of a semiconductor can be direct or indirect. To ensure energy and momentum conservation, the excitation of a valence electron to the conduction band for a indirect bandgap is only possible with an additional phonon absorption or emission process. Electro magnetic waves are able to polarize the lattice of the semiconductor. This leads to a change in the lattice contribution ϵ_{LC} of the dielectric function:

$$\epsilon_{LC} = \epsilon_{\infty} \left(1 + \frac{\omega_{LO}^2 - \omega_{TO}^2}{\omega_{TO}^2 - \omega^2 - i\Gamma\omega} \right) \quad (16)$$

with ϵ_{∞} as the high frequency dielectric constant, ω_{LO} as the longitudinal optical phonon frequency, ω_{TO} as the transversal optical phonon frequency and Γ as the damping constant. Together with eq. ?? the theoretical reflectivity of a semiconductor can be calculated as shown in fig. ??. The theoretical reflectivity shows a minimum at the plasma frequency.

charge carrier density N_{sc}	$1.05 \cdot 10^{24} \text{ m}^{-3}$
effective mass m_{eff}	$0.063 \cdot m_e$
layer thickness d	$500 \mu\text{m}$
damping constant Γ	250 m^{-1}
l.o. phonon frequency ω_{lo}	$\hbar^{-1} 10^{-3} \cdot 36.1 \text{ meV}$
t.o. phonon frequency ω_{to}	$\hbar^{-1} 10^{-3} \cdot 33.2 \text{ meV}$
relative dielectric constant ϵ_{inf}	10.89

Tabelle 2: Constants used for the calculation of the theoretical reflectivity of intrinsic GaAs, shown in fig??. Constants are taken from [?]

For high frequencies the characteristic of the band gap needs to be considered, as inter band excitation processes are possible. For a direct bandgap the

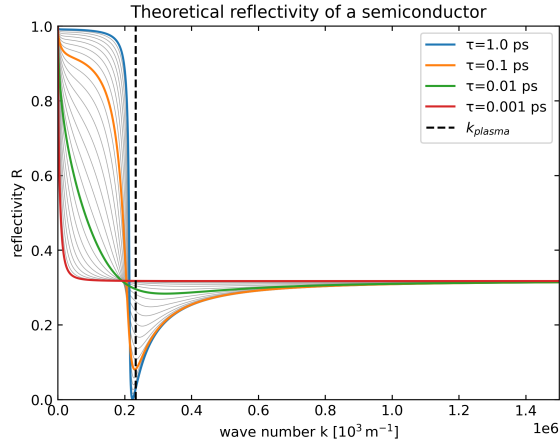


Abbildung 4: Theoretical reflectivity of a semiconductor considering the drude model and polarization effects.

imaginary part of the dielectric function is described with:

$$\epsilon_d'' = \frac{e^2(2\mu)^{\frac{3}{2}}|p_{cv}|^2}{\hbar\epsilon_0 m_e^2 (\hbar\omega)^2} \sqrt{\hbar\omega - E_g} \quad (17)$$

and for an indirect bandgap with:

$$\epsilon_{id}'' \propto (\hbar\omega \pm \hbar\Omega_{ph} - E_g) \quad (18)$$

for $\hbar\omega \geq E_g \pm \hbar\Omega$. $|p_{cv}|$ is the pulse matrix element of the interband transition, μ is the reduced mass of the electron-hole pair, E_g is the bandgap energy, Ω_{ph} is the phonon resonance frequency and m_e is the effective mass of the electron.

3 Set-Up

Fig. ?? shows a simplified schematic version of the *Bruker IFS 66 v/s* fourier spectrometer. The source (1) emits light in the near-infrared range ($1500 \text{ cm}^{-1} - 15000 \text{ cm}^{-1}$) which is then reflected on the beam splitter (3). The Beamsplitter, static mirror (4) and movable mirror (5) can be seen as a michelson interferometer which is used to create an interference pattern. During the measurement, the movable mirror is moved with constant speed. The light is then focused via another mirror (6) on the sample (7). In this case, transmission is shown. For reflection measurements an additional module is used which is not shown in the figure. Mirror (8) then focuses the light onto the detector (9). The signal is then digitalized. Numbers (10) to (13) show the beampath of a laser which is used to determine the position of the movable mirror.

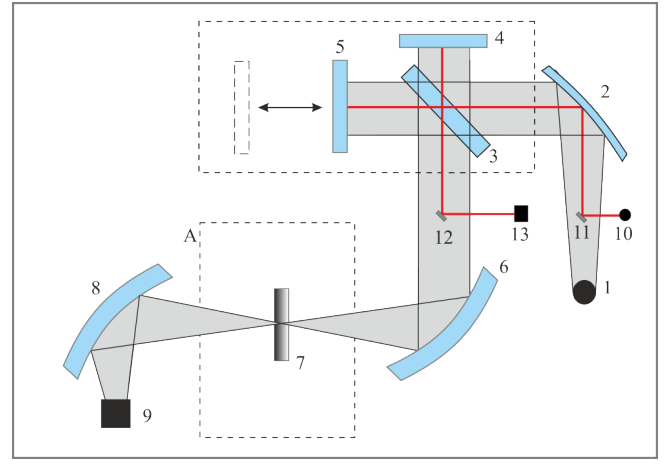


Abbildung 5: Simplified schematic illustration of the set-up used for all experiments. A short overview of the working principle is given in the text.[?]

4 Experiment

All following transmission and reflection experiments are done at ambient air in the chamber. The evaluation of the signal with a 'Mertz' phase correction is done automatically by the system. Furthermore, the spectral range for all measurements is set to: $(1500 - 15000) \text{ cm}^{-1}$.

4.1 Gas absorption

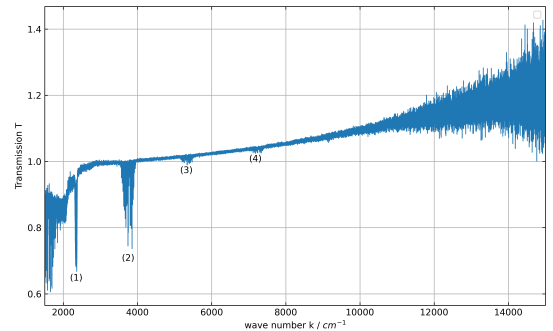


Abbildung 6: Normalized transmission of an empty chamber filled with ambient air. The spectral ranges for gas absorption are highlighted.

To identify the spectral ranges where the result is influenced by the gas absorption, a measurement at ambient air with a resolution of 0.3 cm^{-1} and $N = 10$ scans is performed. The measurement at air is normalized. The corresponding reference measurement is done under vacuum. One expects gas absorption dips at frequencies that are similar to the resonance frequencies of the air molecules. The wavenumbers of

the measured gas absorption is shown in tab ?? All

CO ₂	H ₂ O CO ₂	H ₂ O	H ₂ O
2350 cm ⁻¹	3750 cm ⁻¹	5320 cm ⁻¹	7180 cm ⁻¹

Tabelle 3: Regions of strong gas absorption in fig ??.
The measured values agree with literature values for H₂O and CO₂ [?].

the measured data agrees with the literature values for the gases H₂O and CO₂.

4.2 Signal to noise ratio

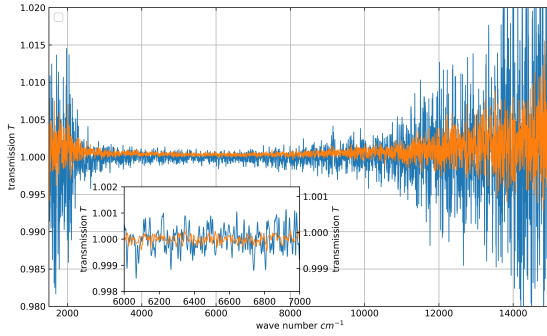


Abbildung 7: Transmission measurements for $N_{\text{scan}} = 10$ and $N_{\text{scan}} = 100$ mirror passes. The inset shows the part used for evaluation of the signal to noise ratio (see fig. ??)

To determine the influence of the number of scans N_{scan} on the signal to noise ratio, measurements with a resolution of 4 cm⁻¹ are done at various N_{scan} . Those are transmission measurements with ambient air. Between 6000 cm⁻¹ and 7000 cm⁻¹ a region with generally less noise (see fig. ??) and no gas absorption (see fig. ??) is identified. To quantitatively examine the behaviour of the signal to noise ratio, the mean value T_{mean} is divided by the Standard deviation T_{std} for every transmission measurement. Thus the signal to noise ratio SNR is given by:

$$SNR = \frac{T_{\text{mean}}}{T_{\text{std}}} \quad (19)$$

Theoretically, one expects SNR to grow according to:

$$SNR = N^a \quad (20)$$

with $a = 0.5$ [?]. To check this dependency, the signal to noise ratio is plotted for various N 's in a double logarithmic scale. From the slope of the linear regression a can be determined as:

$$a = \quad (21)$$

which matches with the expectation.

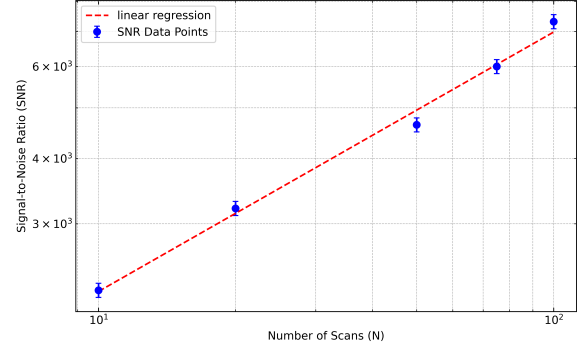


Abbildung 8: SNR over the number of mirror passes in double logarithmic scaling using the spectral range between 6000 cm⁻¹ and 7000 cm⁻¹.

4.3 Transmission at high resolution

4.4 Reflection and transmission at low resolution

4.5 Refractive index N

The complex refractive index is given by eq. ?? with a real and an imaginary part. The absorption coefficient can be determined by numerically solving equations ?? and ??, using the measured reflection and transmission spectra (see sec. ??). The result is shown in fig. ?. The extinction coefficient can be calculated by

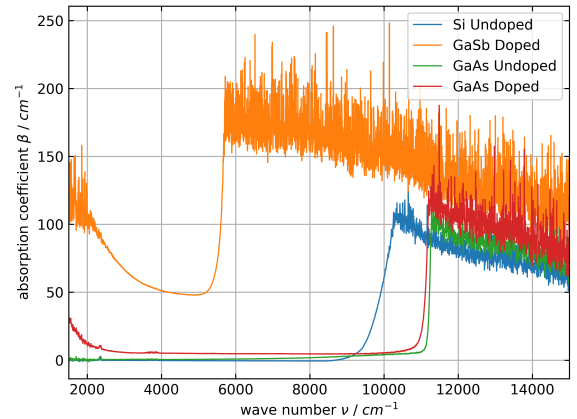


Abbildung 9: Absorption coefficient β of different semiconductor samples.

using eq. ??, shown in fig. ?. With the knowledge of the extinction coefficient κ an eq. ??, the real part of the refractive index n can be calculated. The result is shown in fig. ?. The transmission of the doped Si sample is barely recordable, which is why this sample is not used for the calculation of the optical constants.

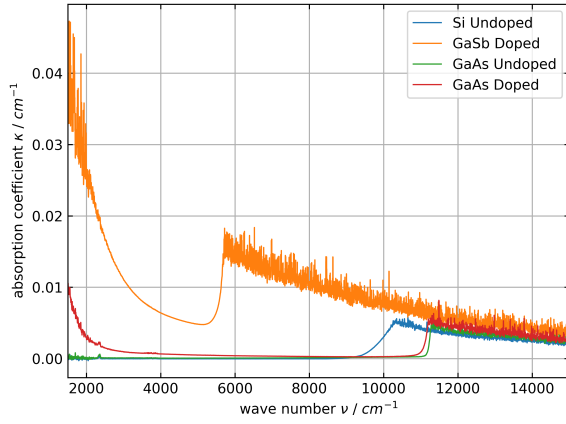


Abbildung 10: Extinction coefficient κ of different semiconductor samples.

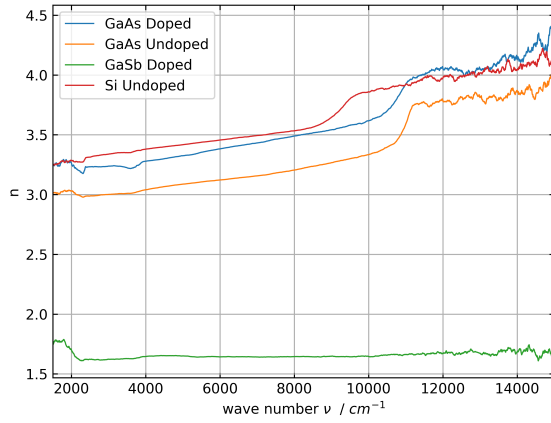


Abbildung 11: Refractive index n of different semiconductor samples.

4.6 Bandgap and pulse matrix element

4.6.1 Direct semiconductor

The band gap E_g and pulse matrix element $|p_{cv}|$ can be evaluated by using eq. ???. When plotting $(\epsilon''^2 \omega^2)^2$ over the wave number k a linear regime around the bandgap can be found. The dependency given in eq. ??? allows the determination of the bandgap by extrapolating the linear regime to $(\epsilon''^2 \omega^2)^2 = 0$ to obtain the bandgap. $|p_{cv}|$ is given by the slope of the extrapolation. The method of obtaining the bandgap and pulse matrix element is shown in fig. ???. The determined values for the given direct semiconductor samples are shown in tab. ???. All errors are calculated with a maximum error estimation for different sizes of the linear regime. To examine the accuracy of the measured data, literature values for the bandgaps of the intrinsic materials are given in tab. ???. The comparison of values is only possible with intrinsic

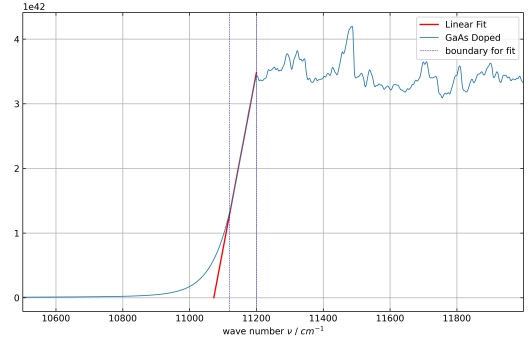


Abbildung 12: Determination of the bandgap and pulse matrix element for a doped GsAs sample.

E_g (GaAs undoped)	$GaAs_{Undoped}$
E_g (GaAs doped)	$GaAs_{Doped}$
E_g (GaSb doped)	$GaSb_{Doped}$
$ p_{cv} $ (GaAs undoped)	$GaAs_{Undoped}$
$ p_{cv} $ (GaAs doped)	$GaAs_{Doped}$
$ p_{cv} $ (GaSb doped)	$GaSb_{Doped}$

Tabelle 4: Values of the bandgap and pulseMatrix-Element for the given direct semiconductors. The data is obtained by extrapolating like described in [?]

materials, because no information about the doping material and concentration is given for the used doped samples. However, the measured values for the bandgap are in good agreement with the literature values considering this.

4.7 Indirect semiconductor

For indirect semiconductors, an additional phonon is required for a electron to be excited form the valence band to the conduction band (see ??). The bandgap can be determined by using eq. ??, by plotting $np.sqrt(\beta\omega^2)$ over the wavenumber. This time, two lineare regimes should be found, which can be seen in the equation. When extrapolating both linear regimes to $np.sqrt(\beta\omega^2) = 0$, intercepts for $E_g - \hbar\Omega_{ph}$ and $E_g + \hbar\Omega_{ph}$ emerge. Adding both intercepts and deviding by two gives the bandgap, whereas the difference of both intercepts divided by two gives the phonon energy $\hbar\Omega_{ph}$. An example for the extrapolation is given in fig. ???. The determined values for the given indirect semiconductor samples are shown in tab. ???. For intrinsic Si one findes in the literature a value of the bandgap of 1.12 eV [?], which is in good agreement with the measured value.

E_g (GaAs undoped)	1.424 eV
E_g (GaSb undoped)	0.726 eV

Tabelle 5: Bandgap energies for intrinsic GaAs and GaSb at room temperature [?].

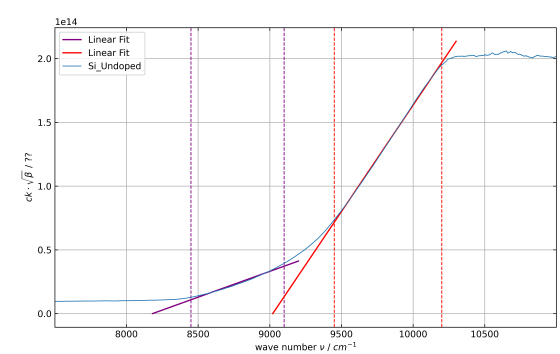


Abbildung 13: Determination of the bandgap and pulse matrix element for an undoped Si sample.

5 Bibliography

Literatur

[1] Meschede D., *Gerthsen Physik*, Springer Verlag, 25th Edition, 2018. Modified by Lukas Hein and Elias Schwarzkopf.

[2] Material for F-Praktikum.??

[3] ?????

[4] Bundesmann C. *Die Bestimmung der optischen Eigenschaften von Halbleitern mit Fourier-Transform-Infrarot-Spektroskopie* <https://polariton.exphysik.uni-leipzig.de/subsites/staff/bundesmann/ftir/ifs48.html>. Accessed on 2025-04-02.

[5] ?????????????????

[6] ???????

[7] Sun, J. et al. (2024) ‘Tunable diode laser absorption spectroscopy for detection of multi-component gas: a review’, *Applied Spectroscopy Reviews*, 59(8), pp. 1086–1107. doi: 10.1080/05704928.2024.2302608.

E_g (Si undoped)	$si_{Undoped}$
$\hbar\Omega_{ph}$ (Si undoped)	$si_{Undoped}$

Tabelle 6: Values of the bandgap and phonon energy for undoped Si. The data is obtained by extrapolating like described in [?]